

Commerce XI Maths App

Exercise 1.3

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 18

Question 1 - Q1(i)

The following table gives frequency distribution of marks of 100 students in an examination.

Marks	15-20	20-25	25-30	30-35	35-40	40-45	45-50
No. of students	9	12	23	31	10	8	7

Determine Q_1 graphically.

Correct answer: 26

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 26.

Question 2 - Q1(ii)

The following table gives frequency distribution of marks of 100 students in an examination.

Marks	15-20	20-25	25-30	30-35	35-40	40-45	45-50
No. of students	9	12	23	31	10	8	7

Determine D_6 graphically.

Correct answer: 32.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 32.5.

Question 3 - Q1(iii)

The following table gives frequency distribution of marks of 100 students in an examination.

Marks	15-20	20-25	25-30	30-35	35-40	40-45	45-50
No. of students	9	12	23	31	10	8	7

Determine P_{85} graphically.

Correct answer: 40

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 40.

Question 4 - Q2

The following table gives the distribution of daily wages of 500 families in a certain city.

Daily wages	Below 100	100–200	200–300	300–400	400–500	500–600	600 and above
No. of families	50	150	180	50	40	20	10

Draw a 'less than' ogive for the above data. Determine the median income and obtain the limits of income of central 50% of the families.

Options

A. 150, 228, 298

B. 150, 228, 300

C. 148, 225, 298

D. 150, 250, 350

Correct answer: 150, 228, 298

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 150, 228, 298.

Question 5 - Q3

From the following distribution, determine median graphically.

Daily wages (in ₹)	Above 300	Above 400	Above 500	Above 600	Above 700	Above 800	Above 900
No. of employees	520	470	399	210	105	45	7

Correct answer: 574

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 574.

Question 6 - Q4(i)

The following frequency distribution shows the profit (in ₹) of shops in a particular area of city.

Profit per shop (in ₹000)	0-10	10-20	20-30	30-40	40-50	50-60
No. of shops	12	18	27	20	17	6

Find graphically the limits of middle 40% shops:

Options

A. 20, 36.5

B. 20, 35

C. 18, 36.5

D. 22, 38

Correct answer: 20, 36.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 20, 36.5.

Question 7 - Q4(ii)

The following frequency distribution shows the profit (in ₹) of shops in a particular area of city.

Profit per shop (in ₹000)	0-10	10-20	20-30	30-40	40-50	50-60
No. of shops	12	18	27	20	17	6

Find graphically the number of shops having profit less than ₹35,000.

Correct answer: 67

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 67.

Question 8 - Q5(i)

The following is frequency distribution of over time (per week) performed by various workers from a certain company.

Overtime (in hours)	Below 8	8-12	12-16	16-20	20-24	24 and above
No. of workers	4	8	16	18	20	14

Determine D_2 graphically.

Correct answer: 13

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 13.

Question 9 - Q5(ii)

The following is frequency distribution of over time (per week) performed by various workers from a certain company.

Overtime (in hours)	Below 8	8-12	12-16	16-20	20-24	24 and above
No. of workers	4	8	16	18	20	14

Determine Q_2 graphically.

Correct answer: 19

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 19.

Question 10 - Q5(iii)

The following is frequency distribution of over time (per week) performed by various workers from a certain company.

Overtime (in hours)	Below 8	8-12	12-16	16-20	20-24	24 and above
No. of workers	4	8	16	18	20	14

Determine P_{61} graphically.

Correct answer: 21.2

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 21.2.

Question 11 - Q6(i)

Draw ogive for the following data and hence find the value of D_1 .

Marks less than	10	20	30	40	50	60	70	80	90
No. of students	4	6	24	46	67	86	96	99	100

Correct answer: 22

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 22.

Question 12 - Q6(ii)

Draw ogive for the following data and hence find the value of Q_1 .

Marks less than	10	20	30	40	50	60	70	80	90
No. of students	4	6	24	46	67	86	96	99	100

Correct answer: 30.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 30.5.

Question 13 - Q6(iii)

Draw ogive for the following data and hence find the value of P_{40} .

Marks less than	10	20	30	40	50	60	70	80	90
No. of students	4	6	24	46	67	86	96	99	100

Correct answer: 37

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 37.

Question 14 - Q7

The following table shows the age distribution of head of the families in a certain country.

Age of head of family (in years)	Under 35	35-45	45-55	55-65	65-75	75 and above
Numbers (million)	46	85	64	75	90	40

Determine the third, fifth and eighth decile of the distribution graphically.

Options

- A. 44, 55.5, 68
- B. 42, 55.5, 70
- C. 44, 57.5, 68
- D. 45, 56, 70

Correct answer: 44, 55.5, 68

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 44, 55.5, 68.

Question 15 - Q8

The following table gives the distribution of females in an Indian village.

Age group	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No. of females (in 000)	175	100	68	48	25	50	23	8	2	1

Determine the median age graphically.

Correct answer: 17.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 17.5.

Question 16 - Q9

Draw ogive for the following distribution and hence find graphically the limits of weight of middle 50% fishes.

Weight of fishes (in gms)	800-890	900-990	1000-1090	1100-1190	1200-1290	1300-1390	1400-1490
No. of fishes	8	16	20	25	40	6	5

Options

A. 1025, 1248

B. 1000, 1250

C. 1025, 1250

D. 1050, 1248

Correct answer: 1025, 1248

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 1025, 1248.

Question 17 - Q10(i)

Find graphically the values of D_3 for the data given below :

I.Q. of students	60-69	70-79	80-89	90-99	100-109	110-119	120-129
No. of students	20	40	50	50	20	10	10

Correct answer: 79.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 79.5.

Question 18 - Q10(ii)

Find graphically the values of P_{65} for the data given below :

I.Q. of students	60-69	70-79	80-89	90-99	100-109	110-119	120-129
No. of students	20	40	50	50	20	10	10

Correct answer: 93.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 93.5.